What is an Incident?

Incident Management & Its Importance

OBJECTIVE

What This Presentation Aims to Achieve

01 Define what an incident is and distinguish it from related concepts such as events, alerts, and problems.

02 Understand the end-to-end incident management process, key roles, and the full incident lifecycle.

03 Recognize why structured incident management matters, for business continuity, technical reliability, and team culture.

04 Learn how alert handling, severity classification, and escalation policies drive faster, more accurate incident response.

AGENDA

01

What is an Incident?

Definition, key concepts, and real-world examples

02

Incident Management

Process, roles, and the incident lifecycle

03

Why It Matters

Business, technical & organizational impact

04

Alert Pipeline Architecture

End-to-end monitoring to resolution flow

05

Severity & Classification

SEV levels, criteria, and response expectations

06

Key Takeaways

Summary and best practices

01 | What is an Incident?

Definition

Any unplanned interruption or degradation in the quality of an IT service that impacts (or has the potential to impact) users, systems, or business operations.

Key Idea: An incident is about IMPACT, not just abnormal behavior.

DB Outage

Production DB crashes → users see errors → Incident

Alert w/o Incident

CPU spikes briefly → no user impact → No Incident

Alert → Incident

Error rate rises → customers affected → Incident declared

Concept Hierarchy

Event
Any observable occurrence

Alert
Notification of abnormal condition

Incident
User-impacting issue

Problem
Underlying root cause

Service Request
User-initiated request (not failure)

02 | What is Incident Management?

Incident management is the structured process of detecting, responding to, resolving, and learning from incidents to restore service quickly and minimize impact.

KEY ROLES

On-call Engineer

First responder; investigates alerts

Incident Commander

Coordinates response, owns decisions

Responders

Engineers actively fixing the issue

Communications Lead

Updates stakeholders & customers

Scribe

Documents timeline and actions

INCIDENT LIFECYCLE

1. Detection
2. Alert Handling
3. Triage
4. Response & Declaration
5. Mitigation & Resolution
6. Recovery
7. Post-Incident Review

03 | Why Incident Management Matters

Business Impact

- Prevents revenue loss from downtime
- Protects brand reputation
- Maintains customer trust & loyalty
- Ensures regulatory compliance

Technical Impact

- Reduces MTTR (Mean Time To Resolve)
- Improves overall system reliability
- Ensures SLA/SLO compliance
- Drives automation & alerting improvements

Organizational Impact

- Builds operational maturity
- Encourages blameless culture
- Improves cross-team coordination
- Reduces on-call engineer burnout

Without IM: Alert ignored → 20 min unnoticed outage | With IM: Actionable alert → mitigated in 3 minutes

04 | Incident Management Architecture

Monitoring & Observability
Metrics · Logs · Traces

Alerting Rules Engine
Thresholds · SLO Alerts · Anomaly Detection

Alert Processing
Deduplication · Correlation · Noise Reduction

Alert Routing
Service Ownership · On-call · Severity

Notification System
Pager · Chat (Slack) · SMS · Email

Incident Management
Creation · War Room · Status Tracking

Escalation Policies
Time-based · Secondary · Management

Resolution & Recovery
Fix/Rollback · Stabilization

Post-Incident Review
Root Cause · Improve Alerts · Prevention

Alert handling is the FRONT DOOR of incident management — poor handling = missed incidents or alert fatigue

04 | Key Alert Handling Concepts

1 Signal vs Noise

✓ Actionable · Urgent · Tied to user impact

✗ Noisy · Frequent · Non-actionable

2 Deduplication

✓ 100 server failures = 1 incident ticket

✗ 100 server failures = 100 pages (burnout!)

3 Correlation

✓ API errors + DB latency → same root issue

✗ Separate unrelated alerts for related problems

4 Routing

✓ Right team · Right person · Right time

✗ Wrong team paged · Alerts lost in noise

5 Escalation

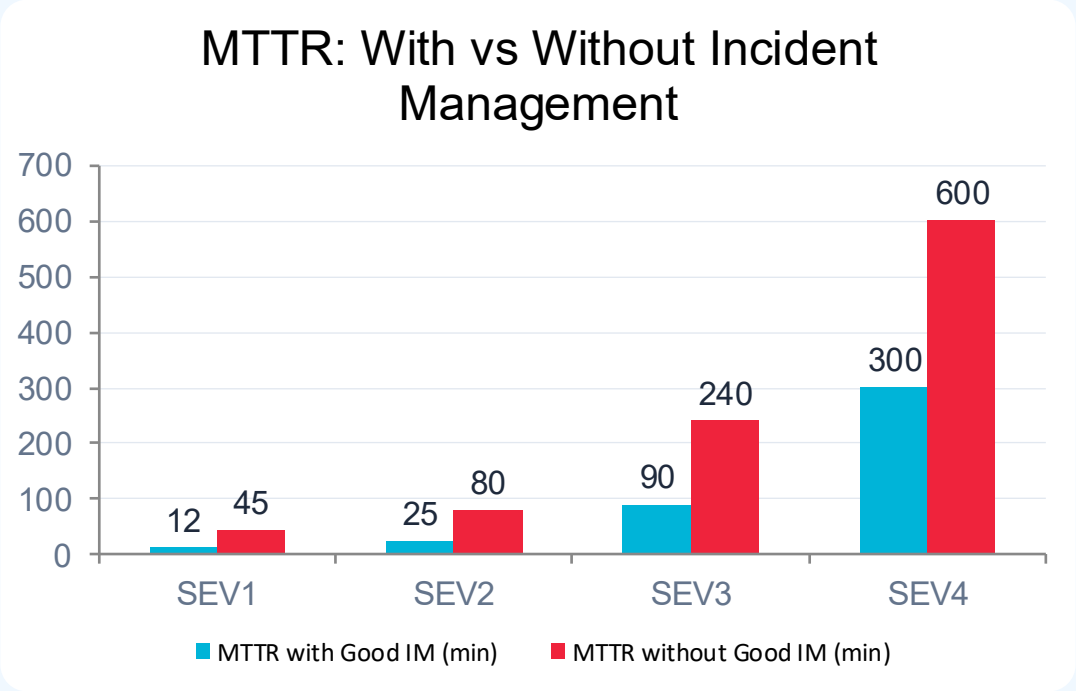
✓ Auto-page secondary on timeout · Notify management

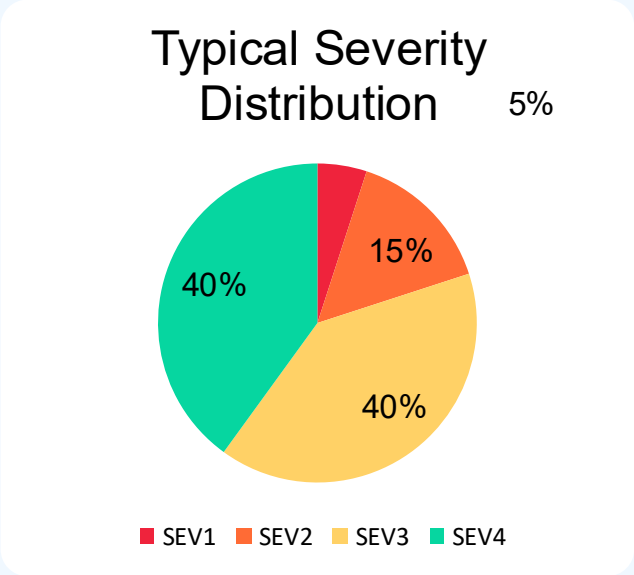
✗ Alert fires → no acknowledgement → ignored

05 | Incident Severity & Classification

Level	Description	User Impact	Response	Alert Behavior
SEV1 / P1	Critical Outage	All users : service unavailable	Immediate (minutes) Full team mobilization	Phone/SMS page · Escalate in minutes
SEV2 / P2	Major Degradation	Significant % : core impaired	Fast (minutes) High-priority response	High-priority alerts · Rapid response
SEV3 / P3	Minor Issue	Small % : workarounds exist	Normal business hours	Ticket or non-urgent alert
SEV4 / P4	Low Priority	Minimal / cosmetic	Scheduled investigation	Logged : not paged

05 | Why Severity Matters: By the Numbers





3 min Avg mitigation with good IM

20 min Avg mitigation without good IM

83% Reduction in customer impact

90%+ Alert noise is non-actionable

KEY TAKEAWAYS

Incidents = Impact

An incident is defined by its impact on users/business, not just abnormal system behavior.

Structured Response is Critical

Clear roles (IC, Responders, Scribe) and a defined lifecycle ensure fast, coordinated resolution.

Alerts Are the Front Door

Poor alert handling leads to fatigue and missed incidents. Good alerts are actionable and routed correctly.

Measure to Improve

Track MTTR, SLO compliance, and incident frequency to continuously reduce impact and recurrence.

Learn from Every Incident

Post-incident reviews drive blameless culture, better tooling, and long-term system resilience.

Speed + Correctness = Success

The best incident management combines fast detection with accurate response to minimize customer impact.